

Statement of Work: Snowflake Crime Data ETL Pipeline

Project Name: Snowflake Crime Data ETL Pipeline

Date: 20 July, 2026

Introduction & Objectives

The objective of this project is to design and implement a scalable Python-based data ETL pipeline in Snowflake that transforms raw crime data into a clean, aggregated, and reporting-ready dataset suitable for use in BI analysis. Particular emphasis should be placed on data quality and validation throughout the pipeline.

Scope of Work

The **Project Manager** is responsible for:

- Documentation - Statement of Work, README, Workflow Diagram, Presentation
- Project oversight + ensure logical workflow
- Set up & Manage GitHub repository

The **Data Ingestion Engineer** is responsible for:

- Data Ingestion

The **Data Quality Engineer** is responsible for:

- Data Cleaning & Validation

The **Analytics & BI Engineer** is responsible for:

- Data Transformation & Aggregation
- Export Final Dataset
- Data Dictionary

Project Deliverables

1. ETL Pipeline (Snowflake - Python Notebook)
2. Reporting Dataset (CSV file)
3. Workflow Diagram
4. Data Dictionary
5. README File

Timeline & Milestones

Milestone	Deliverable	Estimated Completion
Setup & Planning	ETL Pipeline	21 Jul, 2026
Ingestion Layer	ETL Pipeline	27 Jul, 2026
Cleaning & Validation Layer	ETL Pipeline	28 Jul, 2026
Feature Engineering & Transformation Layer	ETL Pipeline	28 Jul, 2026
Aggregation for Reporting	ETL Pipeline	29 Jul, 2026
Export & Final Validation	ETL Pipeline / Reporting Dataset	29 Jul, 2026
Documentation & Deliverables	Workflow Diagram / Data Dictionary / README File	30 Jul, 2026

Datasets Used

- Crime Data:
 - Source: [Data Downloads | data.police.uk](https://data.police.uk/)
- Population Data:
 - Source: [Population estimates for police force areas in England and Wales by single year of age and sex, mid-1991 to mid-2024 - Office for National Statistics](https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/articles/populationestimatesforpoliceforceareasinenglandandwalesbysingleyearofageandsexmid1991tomid2024/1)
- Deprivation Data:
 - Source: [English indices of deprivation 2025 - GOV.UK ...Welsh Index of Multiple Deprivation \(WIMD\) 2025: index and domain ranks and groups, for Lower layer Super Output Areas \(LSOA\) | StatsWales](https://www.gov.uk/government/statistics/english-indices-of-deprivation-2025)

Project Team

Project Manager: Teanna Ottey

Data Ingestion Engineer: Tom Hughes

Data Quality Engineer: Esther James

Analytics & BI Engineer: Noah Esguerra

Signatures

Teanna Ottey | 20th July, 2026

Tom Hughes | 20th July, 2026

Esther James | 20th July, 2026

Noah Esguerra | 20th July, 2026